

Sustainable Olympiad 2026

Round 1: Online Knowledge Challenge

Senior Division (ages 15-18)

Time allowed	90 minutes
Total marks	60
Part A	10 multiple-choice questions, 20 marks
Part B	5 structured problems, 40 marks

Instructions

- Answer **all** questions. Work together as a team.
- Part A: choose **one** answer (A, B, C or D) for each question. There are no negative marks.
- Part B: show all your working. Marks are given for correct method even if the final answer is wrong.
- Give numerical answers to an appropriate number of significant figures, with units.
- Scientific calculators are allowed. Internet access and AI tools are not allowed during the round.
- A data sheet is provided on the next page.

Team name

School

Data sheet

- Acceleration due to gravity: $g = 9.81 \text{ m s}^{-2}$
- Specific heat capacity of water: $c = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$; density of water 1.00 kg L^{-1}
- $1 \text{ kWh} = 3.6 \text{ MJ}$; $0 \text{ }^{\circ}\text{C} = 273.15 \text{ K}$; $1 \text{ year} = 8760 \text{ h}$
- Planck constant $h = 6.63 \times 10^{-34} \text{ J s}$; speed of light $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
- Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
- Molar masses (g mol^{-1}): H 1.008, C 12.01, N 14.01, O 16.00
- Radiative forcing of CO_2 : $\Delta F = 5.35 \ln(C/C_0) \text{ W m}^{-2}$

Part A: Multiple choice (20 marks)

Each question is worth 2 marks. Choose one answer.

- A1.** A rooftop solar module has an area of 1.6 m^2 and an efficiency of 20%. On a day equivalent to 4.5 peak sun hours at 1000 W m^{-2} , approximately how much electrical energy does it produce? **[2]**
- A** 0.32 kWh
B 1.44 kWh
C 7.2 kWh
D 14.4 kWh
- A2.** Over 20 years, the global warming potential (GWP) of methane is about 81, but over 100 years it is only about 28. What is the main reason for this difference? **[2]**
- A** Methane absorbs infrared radiation more strongly as its concentration rises
B Methane has an atmospheric lifetime of only about 12 years, so most of its warming happens in the first decades
C Methane is oxidised to CO_2 , which has a higher GWP than methane
D The 100-year GWP includes absorption of methane by the oceans
- A3.** According to Betz's law, what is the maximum fraction of the kinetic energy flux of the wind that an ideal turbine can extract? **[2]**
- A** $1/3$
B $1/2$
C $16/27$
D $8/9$
- A4.** The power available to a wind turbine is $P = \frac{1}{2}\rho A v^3$. If the mean wind speed at a site is 9.0 m s^{-1} instead of 6.0 m s^{-1} , by what factor does the available power increase? **[2]**
- A** 1.5
B 2.25
C 3.375
D 4.5

- A5.** The average pH of the surface ocean has fallen from about 8.2 to 8.1 since pre-industrial times. By approximately what percentage has the hydrogen-ion concentration increased? [2]
- A** 1%
 - B** 10%
 - C** 26%
 - D** 100%
- A6.** Producers in a grassland fix $50\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$. Assuming 10% of the energy is transferred between successive trophic levels, how much energy is available to tertiary consumers? [2]
- A** $5000\text{ kJ m}^{-2}\text{ yr}^{-1}$
 - B** $500\text{ kJ m}^{-2}\text{ yr}^{-1}$
 - C** $50\text{ kJ m}^{-2}\text{ yr}^{-1}$
 - D** $5\text{ kJ m}^{-2}\text{ yr}^{-1}$
- A7.** In a eutrophic lake, which process is most directly responsible for the sharp fall in dissolved oxygen after an algal bloom? [2]
- A** Algae absorbing oxygen through their cell walls during the day
 - B** Aerobic decomposition of dead algae by bacteria
 - C** Nitrogen fixation by cyanobacteria
 - D** An increase in oxygen solubility as the water cools
- A8.** Which of the following is a **negative** (stabilising) feedback in the climate system? [2]
- A** The ice-albedo feedback
 - B** Release of methane from thawing permafrost
 - C** The water-vapour feedback
 - D** Increased emission of infrared radiation from a warmer surface
- A9.** A geothermal power plant takes heat from a reservoir at $180\text{ }^{\circ}\text{C}$ and rejects heat to cooling water at $30\text{ }^{\circ}\text{C}$. What is its maximum theoretical (Carnot) efficiency? [2]
- A** 17%
 - B** 33%
 - C** 67%
 - D** 83%
- A10.** A heat pump with a coefficient of performance (COP) of 3.5 delivers 14 kW of heat to a building. At what rate does it extract heat from the outside air? [2]
- A** 3.5 kW
 - B** 4.0 kW
 - C** 10 kW
 - D** 49 kW

Part B: Structured problems (40 marks)

B1. Decarbonising school heating (10 marks)

A school needs 108 000 kWh of heat per year. The heat is supplied by a gas boiler with an efficiency of 90%. Burning natural gas releases 0.20 kg CO₂ per kWh of gas burned. The school is considering an air-source heat pump with a seasonal COP of 3.0. Grid electricity has an emission factor of 0.25 kg CO₂ per kWh.

(a) Calculate the annual CO₂ emissions from the gas boiler, in tonnes. [2]

(b) Calculate the annual CO₂ emissions if the heat pump supplies all the heat. [2]

(c) Calculate the percentage reduction in emissions. [2]

(d) Above what grid emission factor would the heat pump cause more CO₂ than the boiler? [2]

(e) Suggest two reasons why the real reduction might differ from your answer to (c). [2]

B2. Fuels and combustion (8 marks)

Model petrol as pure octane, C₈H₁₈ (density 0.74 kg L⁻¹, energy released 44.4 MJ kg⁻¹). Ethanol, C₂H₅OH, releases 26.8 MJ kg⁻¹.

(a) Write a balanced equation for the complete combustion of octane. **[2]**

(b) Calculate the mass of CO_2 produced when 1.00 kg of octane burns completely. **[2]**

(c) A car uses 6.0 L of petrol per 100 km. Calculate its CO_2 emissions in g km^{-1} . **[2]**

(d) Calculate the mass of CO_2 released per MJ of energy for octane and for ethanol. Then explain why bioethanol can still reduce net emissions. **[2]**

B3. Estimating and modelling a population (8 marks)

To estimate a population of ground beetles, students capture 60 beetles, mark them harmlessly and release them. A week later they capture 75 beetles, of which 15 are marked.

(a) Use the Lincoln-Petersen index to estimate the population size. **[2]**

(b) State two assumptions of this method. **[2]**

(c) Chapman's less-biased estimator is $N = (M + 1)(C + 1)/(R + 1) - 1$. **[1]**
Calculate this estimate.

- (d)** The population grows logistically: $dN/dt = rN(1 - N/K)$, with $r = 0.40 \text{ yr}^{-1}$ and $K = 500$. Calculate the growth rate when $N = 300$. Then find the population size at which the growth rate is greatest, and that maximum rate. **[3]**

B4. Rainwater harvesting (7 marks)

A school has a roof area of 450 m^2 , and the annual rainfall is 750 mm. On average, 85% of the rain landing on the roof can be collected. The 600 students each flush a toilet twice per school day, using 6.0 L per flush, on 190 school days a year.

- (a)** Calculate the volume of rainwater collected per year, in m^3 . **[2]**

- (b)** What percentage of the school's toilet-flushing water could rainwater supply? **[2]**

- (c)** All the harvested water is pumped up 12 m to a header tank by a pump that is 50% efficient. Calculate the electrical energy used per year, in kWh. **[3]**

B5. Carbon dioxide and global temperature (7 marks)

The pre-industrial CO_2 concentration was $C_0 = 280 \text{ ppm}$. Assume the equilibrium temperature change is $\Delta T = \lambda \Delta F$, with $\lambda = 0.80 \text{ K per W m}^{-2}$. Ignore all other gases and aerosols.

- (a)** Calculate the radiative forcing and the equilibrium warming for a concentration of 420 ppm. **[2]**

(b) Calculate the equilibrium warming for a doubling of CO_2 (the climate sensitivity in this model). **[1]**

(c) At what CO_2 concentration would the equilibrium warming reach 1.5 K? **[2]**

(d) Observed warming is about 1.3 K, even though CO_2 is above your answer to (c). Give two reasons. **[2]**

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